

VentureMiner AI

Application Flow

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Application Flow

Document 03 — VentureMiner AI End-to-end user and system flows. Where the PRD states what and the TRD states how, this document traces the paths: who clicks what, what happens server-side, what AI agents do, and how state changes.

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1. Purpose & Scope

This document is the map of VentureMiner AI. It covers:

- User journeys from first touch to repeat usage.
- System flows that the user does not see but the engineering team must build.
- Failure paths and recovery.

It does not cover internal coding standards (Document 27) or component design (Document 04).

2. Flow Notation Conventions

- Actor (rectangle) — user, system, agent.
- Decision (diamond) — branching point.
- Action (rounded rectangle) — discrete step.
- State change (cylinder) — persistent artifact.
- Sub-flow (double-bordered rectangle) — referenced flow ID.

3. Primary User Journeys

3.1 First-time user → first useful artifact

1. Land on / (SEO page or paid ad).
2. Sign up via email or OAuth.
3. Land in onboarding (product tour).
4. Pick a "starter track": - Indie Founder → niche exploration → top trend → validation → brief. - Corporate Innovator → adjacency exploration → portfolio → board report. - Investor → thesis portfolio → deal memo. - Consultant → client report template.
5. First artifact generated in ≤ 10 minutes.

Success metric: AC-OB-0001-1 (first-run tour completion), AC-OB-0002-1 (sample opportunity usage), KPI: time-to-first-report (Section 11, PRD).

3.2 Returning user → weekly review

1. Sign in.
2. Land on dashboard with new activity in feed.
3. Review alerts: new trends, fresh evidence on watched opportunities.
4. Drill into a refreshed opportunity.
5. Update score, archive, or add to a report.

3.3 Power user → custom rubric + portfolio

1. Open Settings → Rubric.
2. Create new rubric "Consumer AI", define dimensions, weights.
3. Apply rubric to a portfolio of 20 opportunities.
4. Re-rank; export comparison report.
5. Subscribe to weekly digest on this portfolio.

4. Discovery Flow

4.1 Trigger

- Manual: user clicks "Run Discovery" on a saved search or seed.
- Scheduled: user-defined cron (Daily/Weekly).
- Continuous: always-on mode for Team+ (post-MVP).

4.2 Sequence (manual)



```

participant SR as Source Connectors
participant DB as Postgres
participant OS as OpenSearch

U->>W: Enter seed ("AI for solopreneurs")
W->>DS: POST /discovery {seed, window, depth}
DS->>DB: create discovery_run (status=running)
DS->>AS: enqueue discovery job
AS->>SR: fan out (Trends, Reddit, X, GH, AppStores, G2, HN)
SR-->>AS: raw signals
AS->>AS: cluster, dedupe, score novelty
AS->>DB: persist discovery_hit
AS->>OS: index hits
AS-->>DS: run complete
DS->>W: SSE progress + final result
W->>U: render results, "Pin" CTA per item

```

4.3 Acceptance criteria

- AC-DISC-0001-1: ≥ 20 hits returned for a non-niche seed.
- AC-DISC-0007-1: No duplicate hits across sources.
- AC-DISC-0009-1: Each hit lists its sources.
- AC-DISC-0008-1: Time window is respected.

4.4 Failure modes

- Source timeout $\rightarrow$ marked as partial; user notified; run continues.
- Source error rate $> 50\%$ $\rightarrow$ run aborted, user prompted to disable source.
- LLM error $\rightarrow$ retried with fallback model; final failure surfaces in audit.

5. Validation Flow

5.1 Trigger

- User clicks "Validate" on a discovery hit or saved opportunity.
- User selects depth: Quick (1 source/claim, $< 60s$), Standard (3 sources, 3–8 min), Deep (5+ sources, 8–30 min).

5.2 Sequence (Standard)

```

sequenceDiagram
participant U as User
participant W as Web App
participant VP as Validation Pipeline
participant AR as Agent Runtime
participant RS as Research Agents
participant RAG as RAG Service

```

```

participant SC as Scoring Service
participant DB as Postgres

U->>W: Validate opportunity (Standard)
W->>VP: POST /validation {opp_id, depth=standard}
VP->>DB: create validation_run
VP->>AR: orchestrate(plan)
loop per dimension (market, demand, comp, pricing, persona, WTP, GTM, risk)
AR->>RS: spawn specialist agent
RS->>RAG: retrieve relevant corpus
RAG-->>RS: top-k chunks + citations
RS-->>AR: structured findings w/ citations
AR->>VP: append to validation_step
end
AR->>SC: compute scores (uses current rubric)
SC-->>VP: score + breakdown
VP->>DB: persist run
VP-->>W: SSE progress + final
W->>U: render findings + scores; "Generate Report" CTA

```

5.3 Acceptance criteria

- AC-VAL-0001-1..AC-VAL-0010-1: each dimension's panel rendered.
- AC-VAL-0011-1: depth shown to user; time estimate matches run time ±20%.
- AC-VAL-0013-1: user can attach manual evidence.

5.4 Failure modes

- Agent hallucination detected by verifier → re-run with explicit prompts; if fails twice, mark dimension as unverified.
- RAG retrieval empty → degrade gracefully, surface warning, allow user to attach manual evidence.
- Run aborted → partial findings persisted, run marked partial, user can resume.

6. Scoring Flow

6.1 Default rubric (v1.0)

Market size	20%
Growth	15%
Demand	20%
Buildability	15%
Defensibility	15%
AI-fit	15%

6.2 Sequence

1. Validation run produces evidence (per Section 5).
2. Scoring Service receives evidence and current rubric.
3. For each dimension, an LLM judge produces: - Sub-score 0–10. - Rationale (1–3 sentences). - Confidence (low/med/high).
4. Final score = $\Sigma \text{weight} \times \text{sub-score}$.
5. Stored with full breakdown for audit.

6.3 Custom rubric flow (Team+)

1. User opens Rubric editor.
2. Adds / edits / reorders dimensions and weights (must sum to 100).
3. Saves → new rubric version.
4. Re-scores current portfolio (or selected opportunities).
5. User can view history; scores from old versions remain visible.

6.4 Acceptance criteria

- AC-SCORE-0001-1: default rubric exists.
- AC-SCORE-0002-1: user can add/weight dimensions.
- AC-SCORE-0004-1: each change creates a new version.
- AC-SCORE-0005-1: each dimension score shows rationale + evidence.
- AC-SCORE-0008-1: portfolio view sortable by score.

7. Reporting Flow

7.1 One-page brief

- Trigger: "Generate brief" on an opportunity.
- Output: 1-page PDF.
- Generation: < 60s p75.
- Format: 6 fixed sections (see PRD §7.5).

7.2 Full report (Standard)

sequenceDiagram

participant U as User

participant RS as Reporting Service

participant AR as Agent Runtime

participant RAG as RAG Service

participant S3 as Object Store

U->>RS: Generate full report

RS->>AR: report writer agent (with brief, evidence, score)


```

AR->>RAG: pull deep citations
AR-->>RS: section drafts (cover, exec, market, demand, comp, pricing, persona,
WTP, GTM, risk, score, sources)
RS->>RS: assemble + render charts
RS->>S3: store draft
RS-->>U: notify
U->>U: review, request edits
U->>RS: approve + export
RS->>S3: final PDF/DOCX
RS-->>U: download link

```

7.3 Comparison report

- Trigger: select ≥ 2 opportunities $\rightarrow$ "Compare".
- Output: side-by-side table + 1-page narrative per opportunity.
- Generation: < 3 min p75.

7.4 Acceptance criteria

- AC-RPT-0001-1: brief < 60s p75.
- AC-RPT-0002-1: full report 10–25 pages.
- AC-RPT-0004-1: comparison report 1–2 per opportunity + comparison matrix.
- AC-RPT-0006-1: PDF, DOCX, MD, HTML exports.
- AC-RPT-0010-1: every claim is footnoted.

8. Authentication & Onboarding Flow

8.1 Sign up

```

sequenceDiagram
participant U as User
participant W as Web App
participant A as Auth0
participant AS as Auth Service
participant WS as Workspace Service

U->>W: enter email + password (or OAuth)
W->>A: signup
A-->>W: tokens
W->>AS: sync user record
AS->>WS: create default workspace
WS-->>U: welcome
U->>W: email verification link
W->>A: verify
A-->>U: account active

```

8.2 Sign in

- Email + password (Auth0).
- OAuth (Google, GitHub, Microsoft).
- Magic link (post-MVP).
- MFA challenge (TOTP) for users who enrolled.
- Enterprise SSO via SAML / OIDC (Document 21 for detail).

8.3 Onboarding checklist

- AC-OB-0010-1: tracks 5 actions (sign up, complete tour, run first discovery, run first validation, generate first report).

9. Billing & Subscription Flow

9.1 Sign up → Free tier

- Default on signup.

9.2 Upgrade

```
sequenceDiagram
    participant U as User
    participant W as Web App
    participant BS as Billing Service
    participant ST as Stripe
    participant AS as Auth Service

    U->>W: choose plan
    W->>BS: create checkout session
    BS->>ST: create session
    ST-->>U: hosted checkout
    U->>ST: pay
    ST-->>BS: webhook (subscription.created)
    BS->>AS: update entitlements
    AS-->>U: plan active
```

9.3 Downgrade / cancel

- Self-serve; pro-rated credit issued per Stripe settings.

9.4 Enterprise contract

- Sales-led; contract executes via Stripe sales motion.
- Custom legal (MSA, DPA) attached to workspace.

9.5 Acceptance criteria

- AC-BIL-0008-1: pro-rated upgrade/downgrade.
- AC-BIL-0009-1: tax computed.

10. Workspace & Member Management Flow

10.1 Create workspace

- Trigger: "New workspace" (Solo: disabled; Team: 25; Enterprise: unlimited).

10.2 Invite member

1. Owner/Admin enters email + role.
2. Invite email sent (Postmark).
3. Recipient clicks link → sign in (or sign up) → added to workspace.

10.3 Role change

- Owner/Admin can change role.
- Only Owner can transfer ownership.

10.4 Acceptance criteria

- AC-AUTH-0007-1: roles gate write actions.
- AC-AUTH-0008-1: users can belong to multiple workspaces.
- AC-AUTH-0009-1: every role change is logged.

11. Alert & Notification Flow

11.1 Alert types

- New trend matched watchlist.
- New evidence on watched opportunity.
- Score change beyond threshold.
- Validation run completed.
- Report ready.
- Member joined/left workspace.
- Billing event (card expiring, payment failed).

11.2 Delivery channels

- In-app notification center.
- Email.
- Slack (per workspace).

11.3 Acceptance criteria

- AC-DASH-0004-1: user can configure alert rules.
- AC-INT-0005-1: alerts delivered to Slack.

12. Integration Flows

12.1 Slack

- OAuth install per workspace.
- User picks channel for each alert type.
- Outbound via Slack Web API.

12.2 Notion

- OAuth install; user picks database.
- Opportunity pushed on transition; report pushed on generation.

12.3 REST API

- User mints a token in Settings.
- Scoped token: opportunity:read, opportunity:write, report:read, etc.
- Rate-limited per workspace tier.

12.4 Acceptance criteria

- AC-INT-0001-1: OpenAPI 3.1 spec published.
- AC-INT-0009-1: scoped tokens.
- AC-INT-0010-1: tier-based rate limits.

13. Error & Recovery Flows

13.1 LLM run failure

- Detection: error event from agent runtime.
- Action: retry with backoff; fallback model on second failure; surface to user.
- User action: rerun, contact support, or attach manual evidence.

13.2 Payment failure

- Stripe webhook → billing service → notify user.
- After 3 retries, downgrade to Free at end of period.

13.3 Data export / deletion

- AC-PLAT-0007-1: idempotency keys on writes.

• AC-ADMIN-0008-1: configurable retention

14. Admin & Compliance Flows

14.1 Audit log

- Every write action logged with actor, resource, before/after.
- Filterable, exportable (Enterprise).

14.2 DPA / compliance

- Enterprise admin downloads DPA, SOC 2 report, sub-processor list from admin panel.

14.3 SCIM

- IdP pushes users/groups → auth-svc reconciles → entitlements applied.

15. Appendix

15.1 Flow ID index

F-DISC-1	Discovery — manual
F-DISC-2	Discovery — scheduled
F-VAL-1	Validation — Quick
F-VAL-2	Validation — Standard
F-VAL-3	Validation — Deep
F-SCORE-1	Default scoring
F-SCORE-2	Custom rubric scoring
F-RPT-1	One-page brief
F-RPT-2	Full report
F-RPT-3	Comparison report
F-AUTH-1	Sign up
F-AUTH-2	Sign in
F-BILL-1	Upgrade
F-BILL-2	Cancel
F-WS-1	Create workspace
F-WS-2	Invite member
F-NOT-1	Alert config
F-INT-1	Slack install
F-INT-2	Notion install
F-INT-3	API token mint

15.2 Revision history

v0.5	2026-07-20	Doc Team	All flows drafted
v1.0	2026-07-20	Doc Team	First approved version

End of Document 03 — Application Flow. Detailed screen flows are in Document 04 (UI/UX).